New Wheels Project

Data Querying and Analytics with SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

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Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT COUNT (DISTINCT customer_id) AS Total_customers

FROM  order_t;

SELECT

   c.state,

   COUNT(DISTINCT c.customer_id) AS total_customers

FROM customer_t c

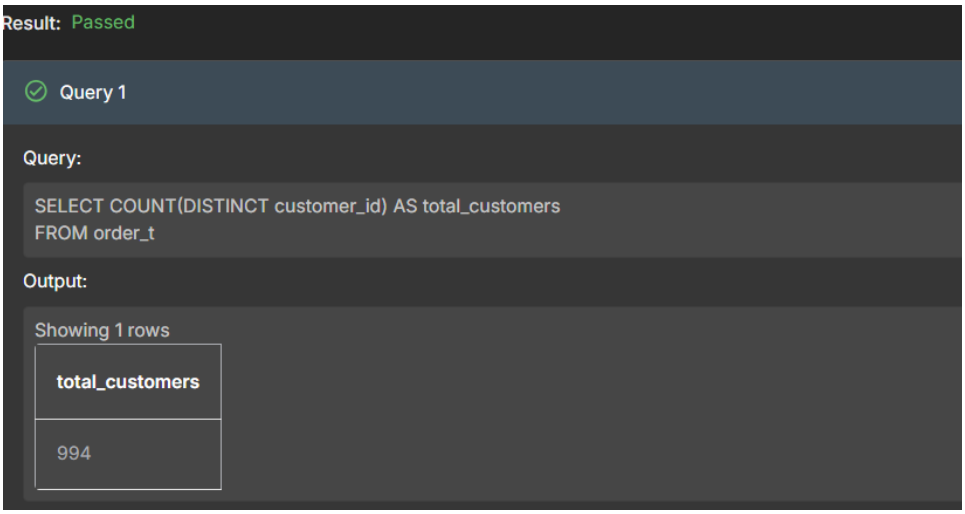
JOIN order_t o

   ON c.customer_id = o.customer_id

GROUP BY c.state

ORDER BY total_customers DESC;
```

Output:



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Result: Passed

Query 1

Query 2

Query:

```
SELECT
   c.state,
   COUNT(DISTINCT c.customer_id) AS total_customers
FROM customer_t c
JOIN order_t o
   ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY total_customers DESC
```

Output:

Showing first 10 rows out of 49 rows

state	total_customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29
Washington	28
Arizona	26

Observations and Insights:

- A total of 994 unique customers has placed orders with New Wheels.
- Customers are distributed across 49 states/regions, indicating a broad geographical presence.
- Texas and California are the top-performing states, each contributing 97 customers.
- Florida (86 customers) and New York (69 customers) also represent strong customer markets.
- Customer concentration is higher in a few states, while several states have relatively fewer customers, highlighting opportunities for regional expansion.

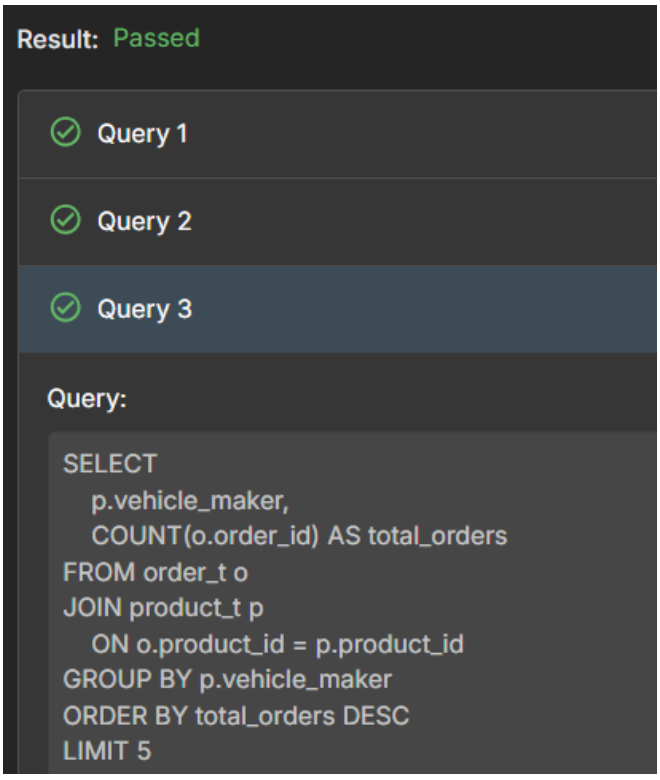
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Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
   p.vehicle_maker,
   COUNT(o.order_id) AS total_orders
FROM order_t o
JOIN product_t p
   ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY total_orders DESC
LIMIT 5;
```

Output:



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Output:

Showing 5 rows

vehicle_maker	total_orders
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Chevrolet is the most preferred vehicle maker, with 83 orders, indicating the highest customer demand among all brands.
- Ford ranks second with 63 orders, followed by Toyota with 52 orders.
- Pontiac and Dodge each received 50 orders, indicating similar customer preference for both brands.
- The top five vehicle makers account for a significant share of customer orders, suggesting that these brands are key contributors to the company's sales.
- Understanding customer preference for these vehicle makers can help New Wheels optimize inventory planning, marketing campaigns, and supplier partnerships.

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Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

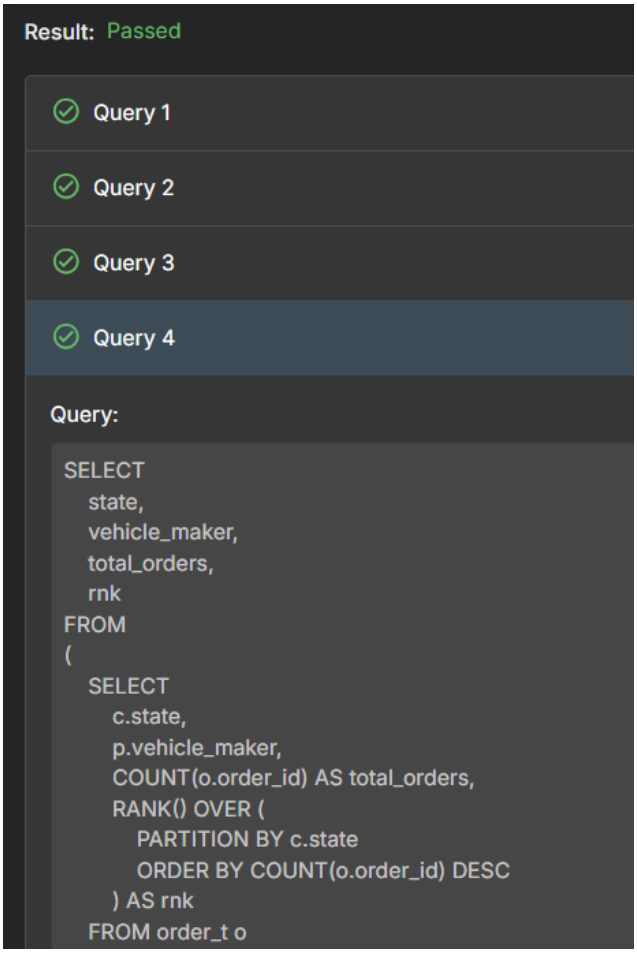
```
SELECT
   state,
   vehicle_maker,
   total_orders,
   rnk
FROM
(
   SELECT
      c.state,
      p.vehicle_maker,
      COUNT(o.order_id) AS total_orders,
      RANK() OVER (
         PARTITION BY c.state
         ORDER BY COUNT(o.order_id) DESC
      ) AS rnk
   FROM order_t o
   JOIN customer_t c
      ON o.customer_id = c.customer_id
   JOIN product_t p
      ON o.product_id = p.product_id
   GROUP BY c.state, p.vehicle_maker
) AS ranked_data
```

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WHERE rnk = 1

ORDER BY state, vehicle_maker;

Output:



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```
    JOIN customer_t c
      ON o.customer_id = c.customer_id
    JOIN product_t p
      ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) AS ranked_data
WHERE rnk = 1
ORDER BY state, vehicle_maker
```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	total_orders	rnk
Alabama	Dodge	5	1
Alaska	Chevrolet	2	1
Arizona	Cadillac	3	1
Arizona	Pontiac	3	1
Arkansas	Chevrolet	1	1

Observations and Insights:

- The most preferred vehicle maker(s) in each state based on the highest number of customer orders.
- In several states, more than one vehicle maker has a rank of 1. This occurs because multiple vehicle makers received the same highest number of orders within that state.
- For example, Arizona has both Pontiac and Cadillac ranked 1, indicating equal customer preference for both brands.
- Similarly, Arkansas has multiple vehicle makers (Volkswagen, Suzuki, Pontiac, Mitsubishi, GMC, and Chevrolet) with Rank = 1, as each received the same number of customer orders.
- The results indicate that while some states have a clear preference for a single vehicle maker, others exhibit diverse customer preferences, resulting in ties for the top rank.

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Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

Overall average rating given by the customers

```
SELECT
   ROUND(AVG(rating),2) AS overall_average_rating
FROM
(
   SELECT
      CASE
         WHEN customer_feedback = 'Very Bad' THEN 1
         WHEN customer_feedback = 'Bad' THEN 2
         WHEN customer_feedback = 'Okay' THEN 3
         WHEN customer_feedback = 'Good' THEN 4
         WHEN customer_feedback = 'Very Good' THEN 5
      END AS rating
   FROM order_t
) AS feedback_rating;
```

Average rating in each quarter

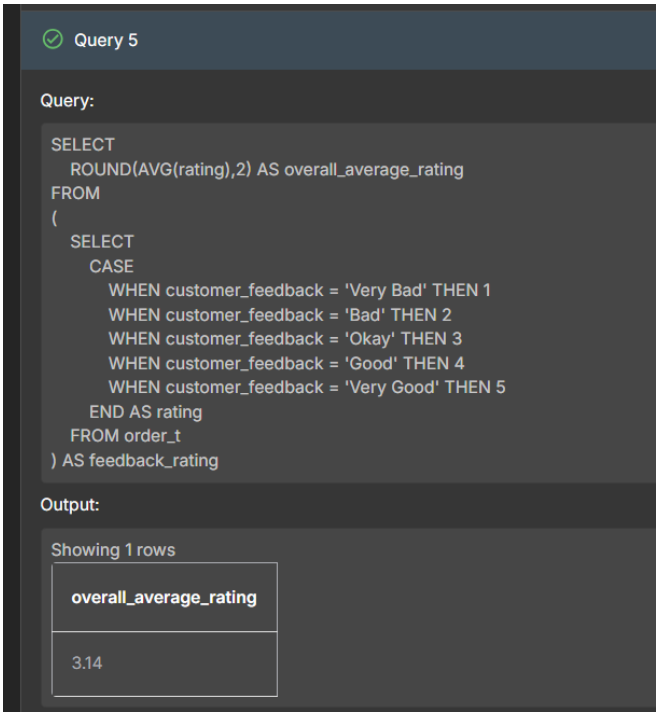
```
SELECT
   quarter_number,
   ROUND(AVG(rating),2) AS average_rating
FROM
(
   SELECT
```

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```
    quarter_number,
    CASE
      WHEN customer_feedback = 'Very Bad' THEN 1
      WHEN customer_feedback = 'Bad' THEN 2
      WHEN customer_feedback = 'Okay' THEN 3
      WHEN customer_feedback = 'Good' THEN 4
      WHEN customer_feedback = 'Very Good' THEN 5
    END AS rating
  FROM order_t
) AS feedback_rating
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Overall average rating given by the customers



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Average rating in each quarter

Query 6

Query:

```
SELECT
   quarter_number,
   ROUND(AVG(rating),2) AS average_rating
FROM
(
   SELECT
      quarter_number,
      CASE
         WHEN customer_feedback = 'Very Bad' THEN 1
         WHEN customer_feedback = 'Bad' THEN 2
         WHEN customer_feedback = 'Okay' THEN 3
         WHEN customer_feedback = 'Good' THEN 4
         WHEN customer_feedback = 'Very Good' THEN 5
      END AS rating
   FROM order_t
) AS feedback_rating
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	average_rating
1	3.55
2	3.35
3	2.96
4	2.4

Observations and Insights:

- The overall average customer rating is 3.14 out of 5, indicating moderate customer satisfaction with New Wheels' products and after-sales services.
- Quarter 1 recorded the highest average rating (3.55), suggesting customers were relatively more satisfied during the beginning of the year.
- Customer satisfaction declined consistently across the quarters, with the average rating decreasing from 3.55 (Q1) to 3.35 (Q2), 2.96 (Q3), and finally 2.40 (Q4).

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- Quarter 4 recorded the lowest average rating, indicating a significant decline in customer experience and service quality towards the end of the year.
- The continuous downward trend in customer ratings aligns with the business problem, which mentions declining customer feedback and a reduction in new customers. This suggests that deteriorating customer satisfaction may be contributing to the company's declining sales.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
  quarter_number,
  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS very_bad_pct,
  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS bad_pct,
  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS okay_pct,
  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS good_pct,
  ROUND(
```

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```
    COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS very_good_pct

FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Query 7

Query:

```
SELECT
  quarter_number,

  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS very_bad_pct,

  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS bad_pct,

  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS okay_pct,

  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0
    / COUNT(*), 2
  ) AS good_pct,

  ROUND(
    COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0
    / COUNT(*), 2
FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number
```

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Output:

Showing 4 rows

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- The percentage of "Very Bad" feedback increased significantly from 10.97% in Q1 to 30.65% in Q4, indicating a substantial rise in customer dissatisfaction.
  Similarly, "Bad" feedback rose from 11.29% in Q1 to 29.15% in Q4, showing that negative customer experiences became more frequent over time.
- The percentage of "Very Good" feedback declined steadily from 30.00% in Q1 to 10.05% in Q4, reflecting a considerable drop in highly satisfied customers.
- "Good" feedback also decreased from 28.71% in Q1 to 10.05% in Q4, further indicating declining customer satisfaction.
- The percentage of "Okay" feedback remained relatively stable across all four quarters, fluctuating between 19% and 22%.

The quarter-wise feedback distribution reveals a consistent shift from positive to negative customer feedback throughout the year. The proportion of 'Very Bad' and 'Bad' feedback increased substantially, while 'Good' and 'Very Good' feedback declined, indicating a deterioration in customer satisfaction over time.

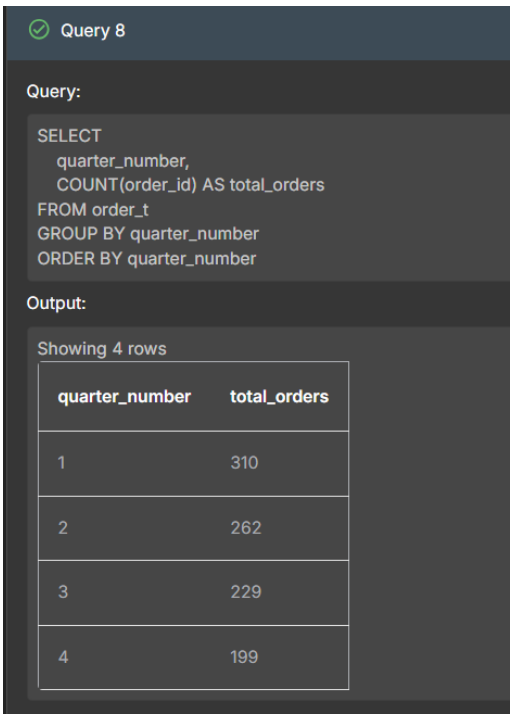
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Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
   quarter_number,
   COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



Observations and Insights:

- The number of customer orders decreased consistently across all four quarters.
- Quarter 1 recorded the highest number of orders (310).
- Orders declined to 262 in Quarter 2, 229 in Quarter 3, and 199 in Quarter 4.
- Overall, the company experienced a steady downward trend in sales volume throughout the year.
- This declining order trend is consistent with the business problem, which mentions a reduction in new customers and declining sales over the past year.

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Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
   quarter_number,
   net_revenue,
   LAG(net_revenue) OVER (ORDER BY quarter_number) AS previous_quarter_revenue,
   ROUND(
      (
         (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
         /
         LAG(net_revenue) OVER (ORDER BY quarter_number)
      ) * 100,
      2
   ) AS qoq_percentage_change
FROM
(
   SELECT
      quarter_number,
      ROUND(
         SUM(
            quantity * vehicle_price * (1 - discount)
       ),
         2
      ) AS net_revenue
```

```
    FROM order_t

    GROUP BY quarter_number

) revenue

ORDER BY quarter_number;
```

Output:

Query 9

Query:

```
SELECT
    quarter_number,
    net_revenue,
    LAG(net_revenue) OVER (ORDER BY quarter_number) AS previous_quarter_revenue,

    ROUND(
        (
            (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
            /
            LAG(net_revenue) OVER (ORDER BY quarter_number)
        ) * 100,
        2
    ) AS qoq_percentage_change

FROM
(
    SELECT
        quarter_number,

        ROUND(
            SUM(
                quantity * vehicle_price * (1 - discount)
            ),
            2
        ) AS net_revenue

    FROM order_t

    GROUP BY quarter_number
) revenue

ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	previous_quarter_reve...	qoq_percentage_change
1	18032549.9		
2	13122995.76	18032549.9	-27.23
3	8882298.84	13122995.76	-32.32
4	8573149.28	8882298.84	-3.48

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Observations and Insights:

- Quarter 1 generated the highest net revenue of 18.03 million, serving as the baseline for quarter-over-quarter comparisons.

  Net revenue declined consistently across all four quarters, indicating a sustained reduction in sales performance.

- Revenue decreased by 27.23% from Q1 to Q2, followed by a larger decline of 32.32% from Q2 to Q3.
- Although revenue continued to decline in Q4, the decrease was relatively smaller at 3.48%, suggesting that the rate of decline slowed toward the end of the year.
- The persistent negative quarter-over-quarter growth aligns with the business problem statement, indicating weakening business performance and highlighting the need to improve customer satisfaction and sales.

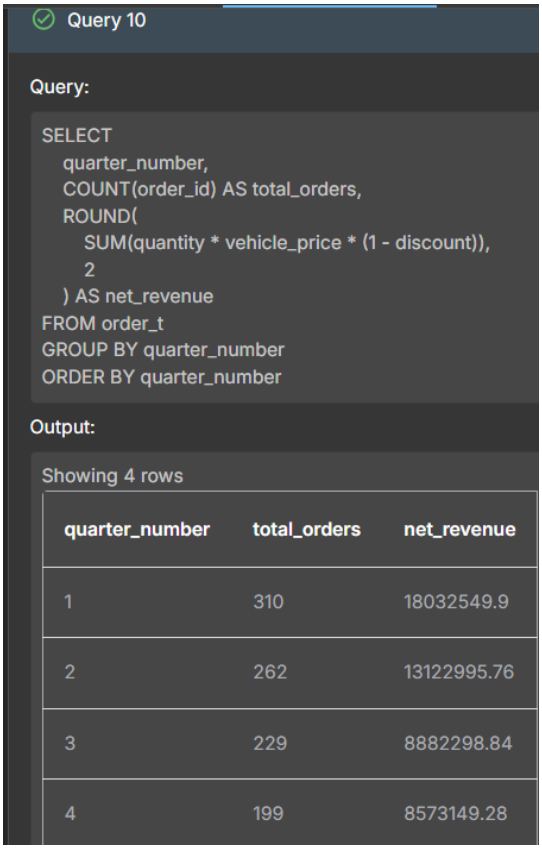
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Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
   quarter_number,
   COUNT(order_id) AS total_orders,
   ROUND(
      SUM(quantity * vehicle_price * (1 - discount)),
      2
   ) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



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Observations and Insights:

- Both total orders and net revenue showed a consistent declining trend across all four quarters.
- Quarter 1 recorded the highest business performance with 310 orders and a net revenue of 18.03 million.
- Total orders decreased from 310 in Q1 to 199 in Q4, while net revenue declined from 18.03 million to 8.57 million during the same period.
- The simultaneous decline in both metrics indicates a positive relationship between order volume and revenue, where fewer customer orders resulted in lower overall revenue.
- The combined trend highlights a continuous slowdown in business performance throughout the year, supporting the business problem statement regarding declining sales.

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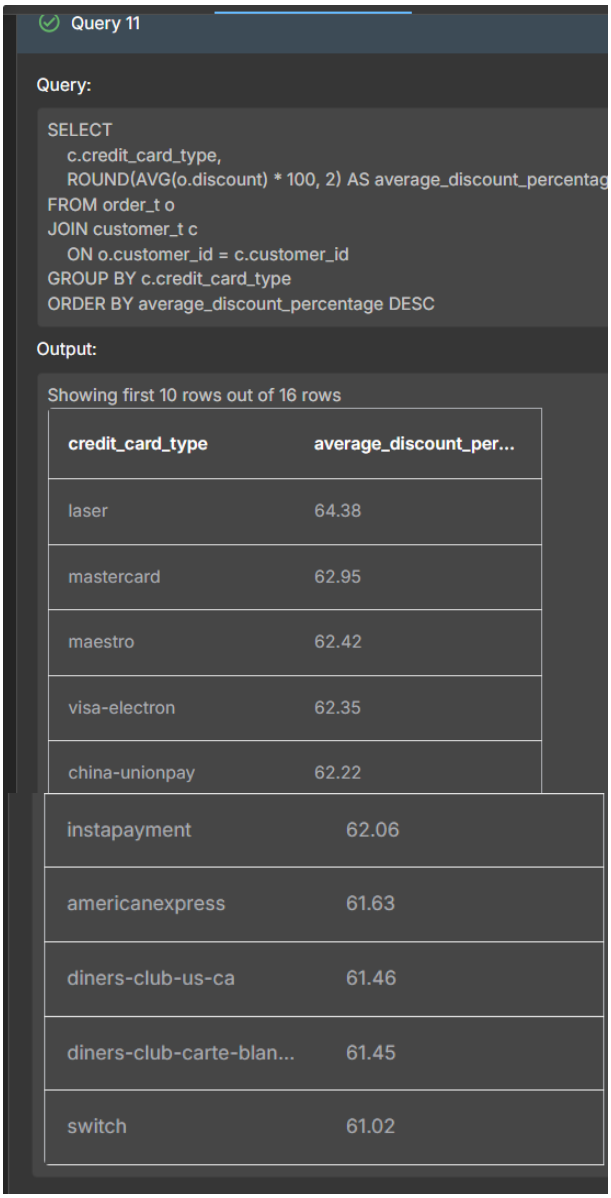
Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
   c.credit_card_type,
   ROUND(AVG(o.discount) * 100, 2) AS average_discount_percentage
FROM order_t o
JOIN customer_t c
   ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY average_discount_percentage DESC;
```

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Output:



Query 11

Query:

```
SELECT
   c.credit_card_type,
   ROUND(AVG(o.discount) * 100, 2) AS average_discount_percentag
FROM order_t o
JOIN customer_t c
   ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY average_discount_percentage DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	average_discount_per...
laser	64.38
mastercard	62.95
maestro	62.42
visa-electron	62.35
china-unionpay	62.22
instapayment	62.06
americanexpress	61.63
diners-club-us-ca	61.46
diners-club-carte-blan...	61.45
switch	61.02

Observations and Insights:

- The average discount offered varies slightly across different credit card types, ranging approximately from 61% to 64%.
  Laser card holders received the highest average discount (64.38%) among all credit card types.
- MasterCard (62.95%), Maestro (62.42%), and Visa Electron (62.35%) also received relatively high average discounts.
- The variation in average discounts across different card types is small, indicating that New Wheels follows a fairly consistent discounting strategy regardless of the payment method.
- Overall, the company does not appear to heavily favor one specific credit card type, as the average discounts remain within a narrow range.

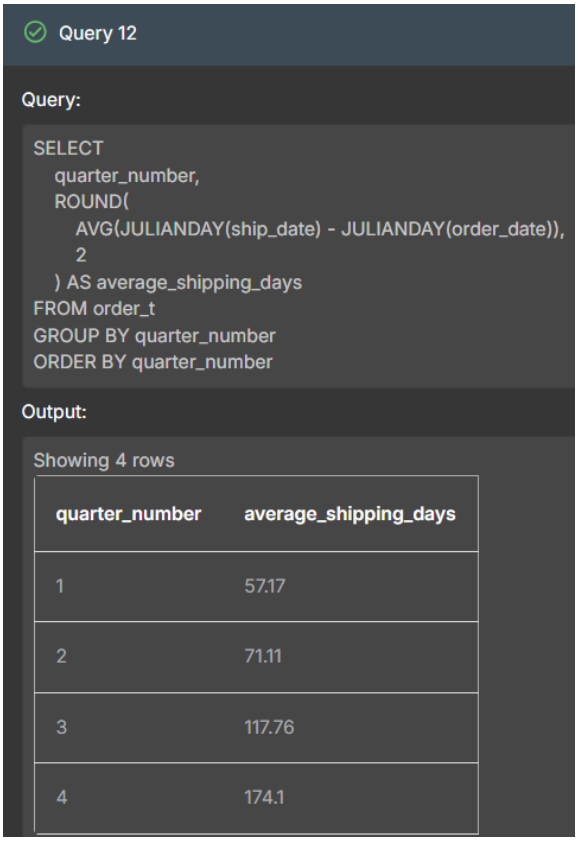
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Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
   quarter_number,
   ROUND(
      AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)),
      2
   ) AS average_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



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Observations and Insights:

- The average shipping time increased steadily across all four quarters, indicating a continuous decline in shipping efficiency.
- Quarter 1 recorded the lowest average shipping time of 57.17 days, suggesting relatively faster order fulfillment.
- The average shipping time increased to 71.11 days in Q2, 117.76 days in Q3, and reached 174.10 days in Q4.
- The significant increase in shipping time may have negatively impacted customer satisfaction, which is consistent with the declining customer ratings observed in previous analyses.
- The increasing delivery time may have contributed to the reduction in customer orders and overall revenue during the year.

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Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
$48,611,993.78	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
8,573,149.28	199	97.96 days	21.50%

Business Recommendations

1. **Improve Shipping and Delivery Efficiency**
   - The average shipping time increased significantly from 57.17 days in Q1 to 174.10 days in Q4, indicating delays in order fulfillment. Optimizing logistics, improving coordination with shipping partners, and reducing delivery lead times can enhance customer satisfaction.
2. **Enhance Customer Satisfaction and After-Sales Service**
   - Customer ratings declined steadily across the quarters, with the average rating dropping from 3.55 in Q1 to 2.40 in Q4. The company should identify common customer complaints, strengthen after-sales support, and implement regular customer feedback reviews to improve service quality.
3. **Increase Customer Retention and Sales**
   - Both orders and net revenue declined consistently throughout the year. Introducing customer loyalty programs, seasonal promotions, personalized offers, and targeted marketing campaigns can help attract new customers and retain existing ones.
4. **Review Discount Strategy**
   - Although discounts were offered across all credit card types, sales continued to decline. The company should evaluate whether its discount strategy effectively drives purchases and consider targeted promotions instead of uniform discounts.
5. **Focus on High-Demand Vehicle Makers**

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- Vehicle preferences differ by state, with brands such as Chevrolet, Ford, and Toyota showing higher demand in several regions. The company should optimize inventory and marketing based on regional customer preferences to maximize sales opportunities.

6. **Monitor Quarterly Business Performance**

- The CEO should regularly monitor key performance indicators (KPIs) such as revenue, order volume, customer ratings, shipping time, and customer feedback through a quarterly dashboard. This will enable early identification of performance issues and support timely business decisions.

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